

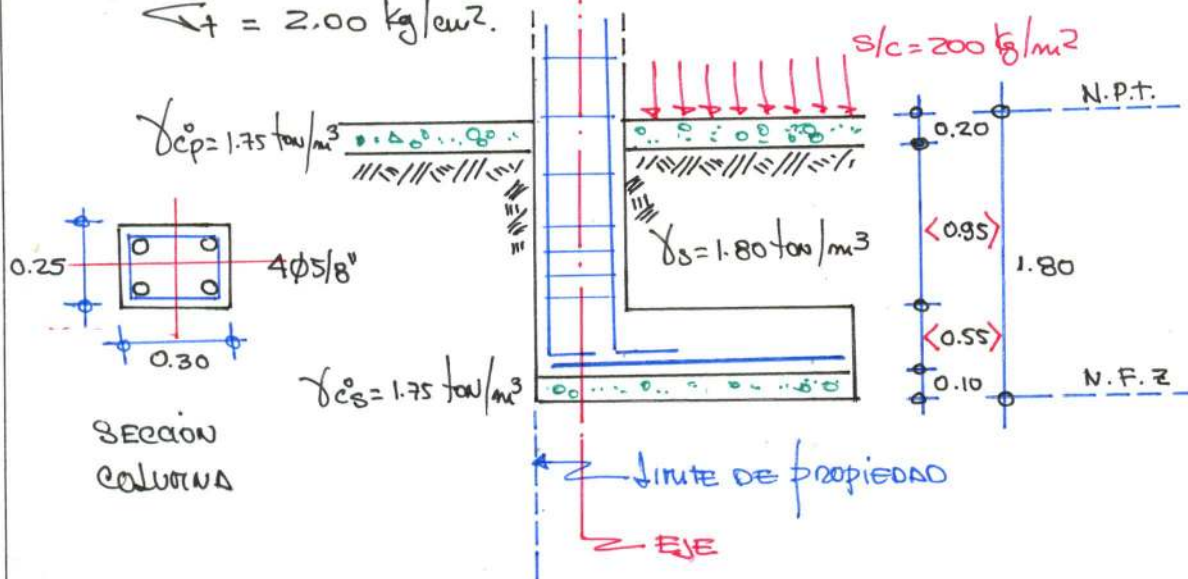
PROBLEMA N° 05 :

DIMENSIONAR, VERIFICAR, DISEÑAR LA ZAPATA EXCÉNTRICA DE LA FIGURA ADJUNTA, PARA LAS SOLICITACIONES QUE SE INDICAN.

$$P_{CH} = 32 \text{ ton} \quad M_{CH} = 6.00 \text{ ton-m} \quad f_c = 210 \text{ kg/cm}^2$$

$$P_{CV} = 12 \text{ ton} \quad M_{CV} = 4.00 \text{ ton-m} \quad f_y = 4200 \text{ kg/cm}^2$$

$$\alpha = 2.00 \text{ kg/cm}^2$$

DIMENSIONAMIENTO :

- PERALTE : $A_{scol} = 4\phi 5/8 \Rightarrow \phi 5/8 \Rightarrow d_b = 1.588 \text{ cm}, A_b = 2.00 \text{ cm}^2$
CONSIDERACION NORMATIVAS

$$l_d = 0.08 d_b f_y / \sqrt{f_c} \Rightarrow l_d = 36.82 \text{ cm} \quad l_d = 36.82 \text{ cm}$$

$$l_d = 0.004 d_b f_y \Rightarrow l_d = 26.678 \text{ cm} \Rightarrow l_d = 35 \text{ cm.}$$

$$l_d \geq 20 \text{ cm.} \Rightarrow l_d = 20 \text{ cm.}$$

$$\therefore d = l_d + 10 \text{ cm} \Rightarrow d = 45 \text{ cm} \Rightarrow h = d + 10 \text{ cm} \Rightarrow h = 55 \text{ cm.}$$

- CAPACIDAD PORTANTE NETA DEL TERRENO (q_e)

$$q_e = \gamma - (\gamma_{cs} \times h_s) - (\gamma_{cz} \times h_z) - (\gamma_s \times h_s) - (\gamma_{cp} \times h_p) - s/c$$

$$q_e = 20.00 - (1.75 \times 0.10) - (2.40 \times 0.55) - (1.80 \times 0.95) - (1.75 \times 0.20) - 0.20$$

$$q_e = 16.245 \text{ ton/m}^2 = 1.6245 \text{ kg/cm}^2$$

SOLICITACIONES :

→ CARGAS DE SERVICIO : $P_s = P_{cm} + P_{cv} \Rightarrow P_s = 32 + 12 \Rightarrow P_s = 44.00 \text{ ton.}$
 $M_s = M_{cm} + M_{cv} \Rightarrow M_s = 6 + 4 \Rightarrow M_s = 10.00 \text{ ton-m.}$

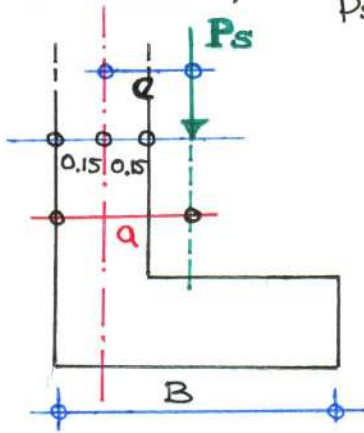
→ CARGAS ULTIMAS : $P_u = 1.40 P_{cm} + 1.70 P_{cv} \Rightarrow P_u = 1.40(32) + 1.70(12) \Rightarrow P_u = 65.20 \text{ ton}$
 $M_u = 1.40 M_{cm} + 1.70 M_{cv} \Rightarrow M_u = 1.40(6) + 1.70(4) \Rightarrow M_u = 15.20 \text{ ton-m.}$

ZAPATAS SIN EXCENTRICIDAD

$$\bar{A} = \frac{P_s}{q_e} \Rightarrow \bar{A} = \frac{44.00}{1.625} \Rightarrow \bar{A} = 2.709 \text{ m}^2$$

∴ TENDER ZAPATA EXCENTRICIDAD $\bar{A} > 2.709 \text{ m}^2$

EXCENTRICIDAD $\Rightarrow e = \frac{M_s}{P_s} \Rightarrow e = \frac{10.00}{44.00} \Rightarrow e = 0.227 \text{ m.}$



$$B_{\min} = 3e \Rightarrow B_{\min} = 0.681 \text{ m}$$

$$B_{\max} = 2a \Rightarrow B_{\max} = 0.754 \text{ m.}$$

$$BL = A : L = 2B : B = \sqrt{\frac{A}{2}}$$

$$B = \sqrt{\frac{2.709}{2}} \Rightarrow B = 1.164 \sim B = 1.20 \text{ m.}$$

∴ $B = 1.20, L = 2.40 \Rightarrow \bar{A} = 2.88 \text{ m}^2$

VERIFICACIÓN DE PRESIÓN $\sigma_{\max} < \sigma_t$

$$\left. \begin{array}{l} B = 1.20 \text{ m} \\ L = 2.40 \text{ m} \\ \bar{A} = 2.88 \text{ m}^2 \\ I = 0.346 \text{ m}^4 \\ c = 0.60 \text{ m.} \end{array} \right\}$$

$$\sigma_{\max} = \frac{P_s}{A} + \frac{M_s c}{I} = \frac{44.00}{2.88} + \frac{10(0.60)}{0.346} \Rightarrow \sigma_{\max} = 32.619 \text{ ton/m}^2 > \sigma_t$$

(NO CUMPLE)
REDIMENSIONAR AREA

$$\left. \begin{array}{l} B = 1.50 \text{ m} \\ L = 3.00 \text{ m} \\ \bar{A} = 4.50 \text{ m}^2 \\ I = 0.844 \\ c = 0.75 \text{ m.} \end{array} \right\}$$

$$\sigma_{\max} = \frac{44.00}{4.50} + \frac{10(0.75)}{0.844} \Rightarrow \sigma_{\max} = 18.664 < \sigma_t$$

(CONFORME)

CARGAS DE DISEÑO (P_u, M_u) $P_u = 65.20 \text{ ton}$ $M_u = 15.20 \text{ ton-m}$.

Reacción Amplificada del Suelo:

$$Q_u = M_u / P_u \Rightarrow Q_u = 15.20 / 65.20 \Rightarrow Q_u = 0.233 < B/6 = 0.25 \text{ m}$$

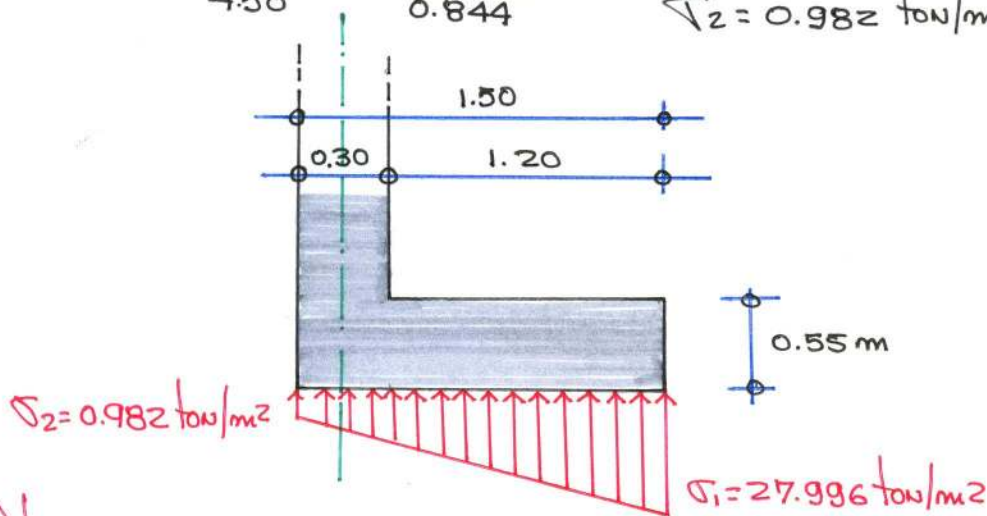
Presión de Contacto F/Trapezoidal

$$\sigma_{1,2} = \frac{P_u}{A} \pm \frac{M_u(c)}{I}$$

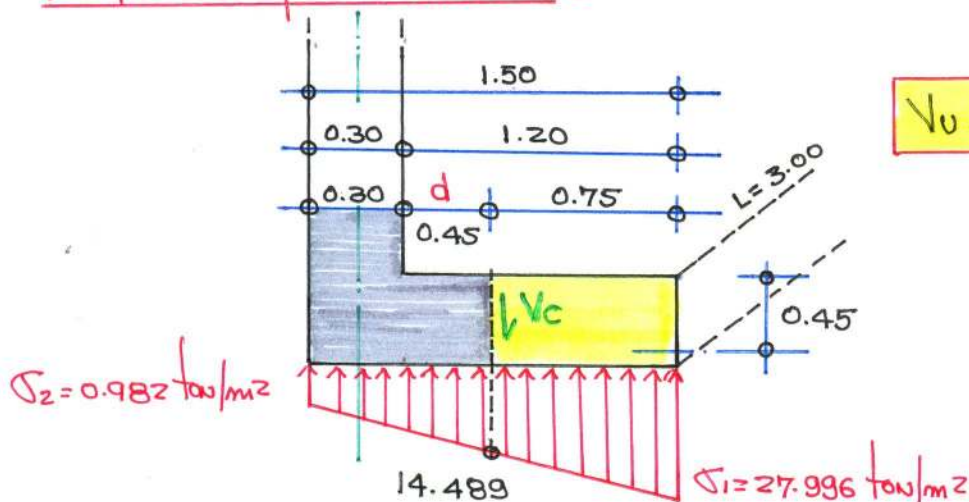
$$\sigma_{1,2} = \frac{65.20}{4.50} \pm \frac{15.20(0.75)}{0.844}$$

$$\sigma_1 = 27.996 \text{ ton/m}^2$$

$$\sigma_2 = 0.982 \text{ ton/m}^2$$



VERIFICACION POR CORTANTE:

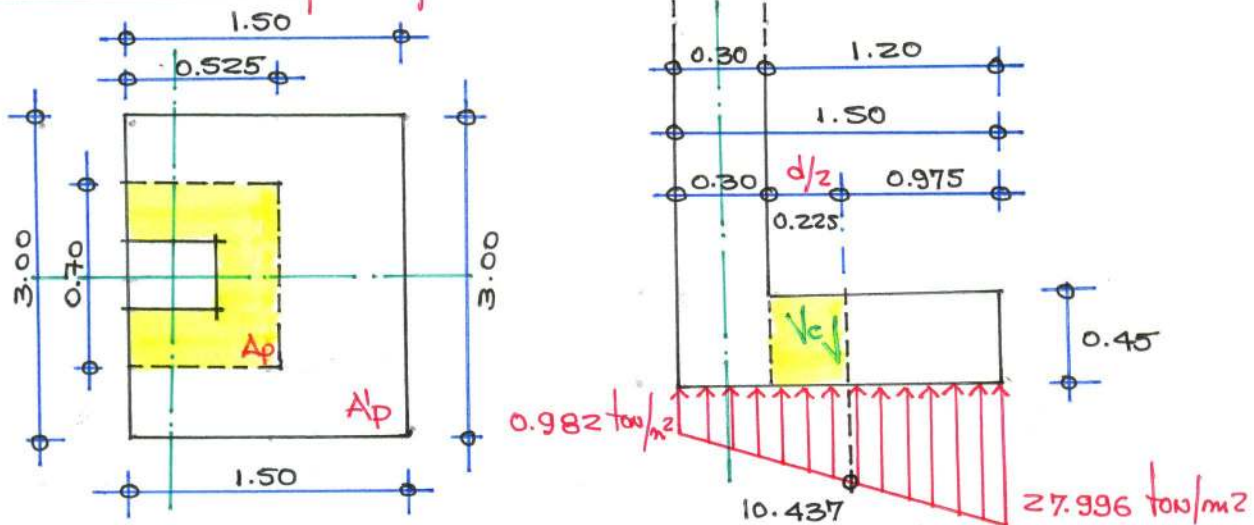


$$V_u \leq \phi V_c$$

$$V_u = \left[\frac{(27.996 + 14.489) \cdot 0.75 \cdot (3.00)}{2} \right] \Rightarrow V_u = 47.796 \text{ ton} = 47795.625 \text{ kg}$$

$$\phi V_c = \phi \cdot 0.53 \sqrt{f_c} \cdot L \cdot d \Rightarrow \phi V_c = 0.85 \cdot 0.53 \sqrt{210} \cdot 300 \cdot 45 \Rightarrow \phi V_c = 88.133 \text{ ton}$$

$$\therefore V_u = 47.796 \text{ ton} < \phi V_c = 88.133 \Rightarrow \text{(CONFORME)}$$

VERIFICACION POR PUNZONAMIENTO:

PERIMETRO DE PUNZONAMIENTO

$$b_0 = 2(0.525) + 0.70 \Rightarrow b_0 = 1.75 \text{ m.}$$

RELACION DE LADOS DE LA COLUMNA

$$\beta_0 = \frac{\text{LADO MAYOR}}{\text{LADO MENOR}} \Rightarrow \beta_0 = \frac{30}{25} \Rightarrow \beta_0 = 1.20$$

$$V_u \leq \phi V_c$$

$$V_u = \left[\frac{(27.996 + 0.982) 1.50 (3.00)}{2} \right] - \left[\frac{(10.437 + 0.982) 0.525 (0.70)}{2} \right]$$

$$V_u = 63.102 \text{ ton} = 63\,102.259 \text{ kg}$$

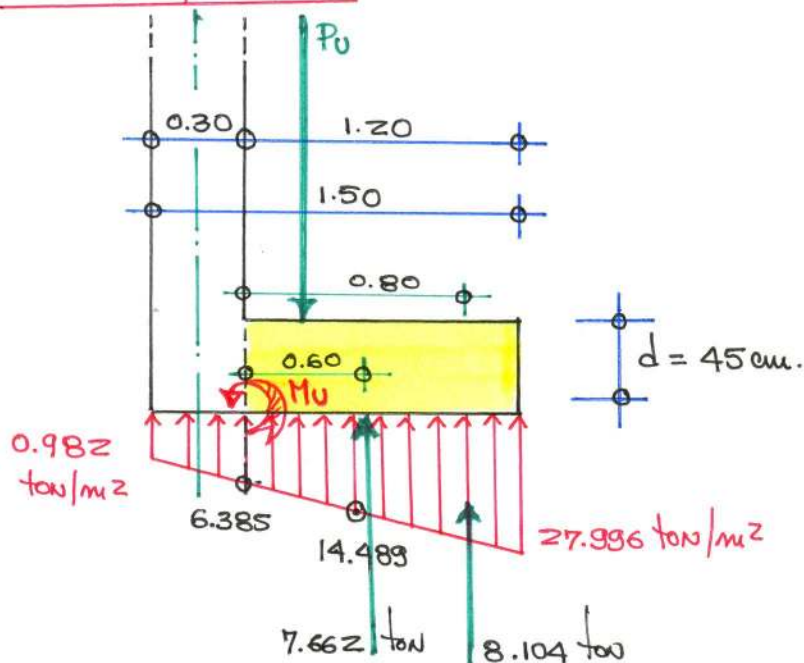
$$\phi V_c = \phi \left(0.53 + \frac{1.10}{\beta_0} \right) \sqrt{f_c} b_0 d$$

$$= 0.85 \left(0.53 + \frac{1.10}{1.20} \right) \sqrt{210} \times 1.75 \times 45$$

$$\phi V_c = 140\,329.058 \text{ kg} = 140.329 \text{ ton.}$$

$$\phi V_c = 140.329 \text{ ton} > V_u = 63.102 \text{ ton}$$

(CONFORME)

DISEÑO DE REFUERZODISEÑO DE REFUERZO LONGITUDINAL

$$w = 11.080 \text{ ton-m}, b = 100 \text{ cm}, d = 45 \text{ cm}, f_c = 210 \text{ kg/cm}^2, f_y = 4200 \text{ kg/cm}^2$$

$$\rho_{min} = 0.70 \sqrt{f_c} / f_y \Rightarrow \rho_{min} = 0.00242 \Rightarrow A_{smin} = \rho_{min} b d \Rightarrow A_{smin} = 10.868 \text{ cm}^2$$

$$\rho_b = 0.85 \beta \frac{f_c}{f_y} \left(\frac{6000}{6000 + f_y} \right) \Rightarrow \rho_b = 0.02125 \Rightarrow A_{sb} = \rho_b b d \Rightarrow A_{sb} = 95.625 \text{ cm}^2$$

$$\rho_{max} = 0.75 \rho_b \Rightarrow \rho_{max} = 0.01594 \Rightarrow A_{smax} = \rho_{max} b d \Rightarrow A_{smax} = 71.719 \text{ cm}^2$$

$$\rho_{dis} = w \frac{f_c}{f_y} \Rightarrow \rho_d = 0.00147 \Rightarrow A_{sd} = \rho_d b d \Rightarrow A_{sd} = 6.629 \text{ cm}^2$$

$$\# \text{ barras } N = \frac{A_{st}}{A_{s\phi}} \quad \therefore \phi 5/8 = 2.00 \text{ cm}^2 \quad A_{st} = 32.604 \text{ cm}^2$$

$$N = \frac{32.604}{2.00} \Rightarrow N = 16.302 \sim \underline{N = 16 \text{ UNIDADES.}}$$

Distribución:

$$S = \frac{3.00 - 0.15 - 0.01588 - 0.10}{15} \Rightarrow \underline{S = 0.18 \text{ m.}}$$

$$\underline{16 \phi 5/8 @ 0.18}$$

DISEÑO DE REFUERZO TRANSVERSA

$$M_w = \frac{14.489 (1.375)^2}{2} \Rightarrow M_w = 13.697 \text{ ton-m.}$$

$$b = 100 \text{ cm}, d = 45.00 \text{ cm}, f_c = 210 \text{ kg/cm}^2, f_y = 4200 \text{ kg/cm}^2$$

$$\rho_{\min} = 0.00242$$

$$A_{st\min} = 10.868 \text{ cm}^2$$

$$\rho_b = 0.02125$$

$$A_{sb} = 95.625 \text{ cm}^2$$

$$\rho_{\max} = 0.01594$$

$$A_{s\max} = 71.718 \text{ cm}^2$$

$$\rho_{dis} = 0.00183$$

$$A_{sd} = 8.23 \text{ cm}^2$$

∴ Acero de Diseño Transversal $A_{sd} = 10.868 \text{ cm}^2$

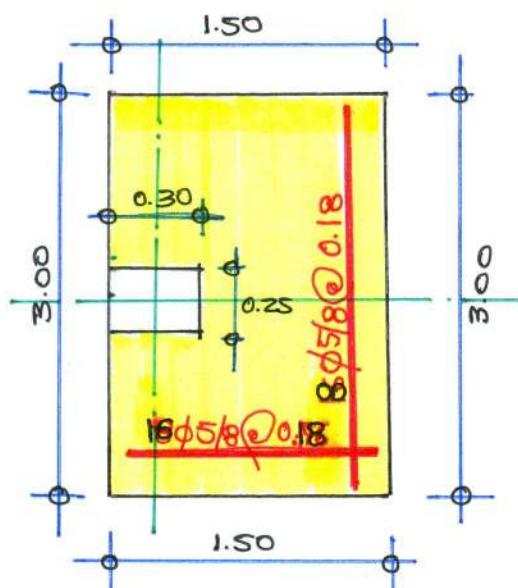
DE ACEROS : $A_{st} = 10.868 (1.50) \Rightarrow A_{std} = 16.302$ $A_{sq \ 5/8} = 2.00$

$$N = \frac{A_{std}}{A_{sq}} \Rightarrow N = \frac{16.302}{2.00} \Rightarrow N = 8.15 \sim N = 8 \text{ UNIDADES}$$

① DISTRIBUCION DE ACEROS (s)

$$S = \frac{1.50 - 0.15 - 0.01588 - 0.10}{7} \Rightarrow S = 0.176 \text{ m.}$$

Acero Transversal : $8 \phi 5/8 @ 0.176 \text{ m.} \sim 8 \phi 5/8 @ 0.18$



DETALLE : EN PLANTA